

Lecture 1: Set and Number System

Sets: Collection of well defined set.

e.g. Symbols, shape.

→ collection of unordered object eg. faults.

Elements: object in a set.

Example: $\{ \underline{1}, \underline{2}, \underline{3}, \underline{4}, \underline{5} \}$

Methods:-

1. Roster Methods $\{1, 2, 3, 4, \dots\}$ (ellipses, ...)

2. Set builder methods. $\{x/x < 6 \text{ and } x \text{ is counting number}\}$

Subset:- $A \subseteq B$

$A = \{1, 2, 3, 4\}$ $B = \{0, 1, 2, 3, 4, 5, 6\}$

Empty set: $\{ \}$ or ϕ

Universal set: denoted by $\rightarrow U$

set containing all elements of which all other set are subset.

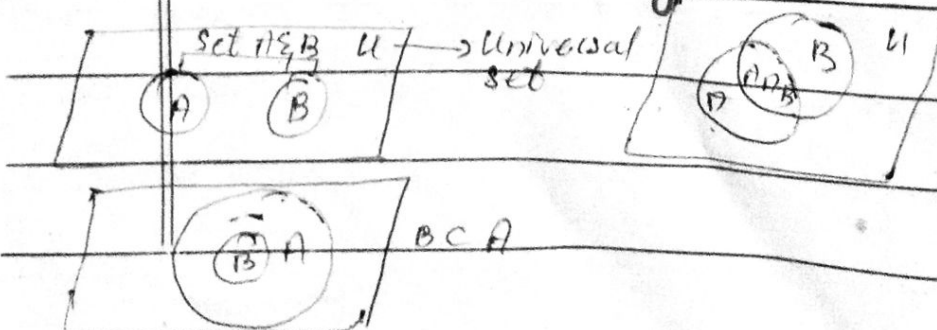
Union:- to take write all numbers in single set.

$A = \{x/x > 5\}$ $B = \{x/x < 3\}$

$A \cup B = \{0, 1, 2, 6, 7, 8\} = \{x/x < 3 \text{ or } x > 5\}$

Intersection: write common no.

Lecture: 2 Venn Diagram:-



- Union and intersection are commutative properties

$$A \cup B = B \cup A \quad | \quad A \cap B = B \cap A$$

$$a \cup b = b \cup a \quad | \quad a \cap b = b \cap a$$

- Union and intersection are associative properties

$$(A \cup B) \cup C = A \cup (B \cup C) \quad | \quad (A \cap B) \cap C = A \cap (B \cap C)$$

$$(a \cup b) \cup c = a \cup (b \cup c) \quad | \quad (a \cap b) \cap c = a \cap (b \cap c)$$

- Distributive laws: - $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
 $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

- Elementary Property:

$$A \cup \phi = A, A \cup A = A \quad | \quad A \cap \phi = \phi, A \cap A = A$$

- Whole numbers (W), Natural numbers (N), Integers ($\mathbb{Z}, \mathbb{Z}^+, \mathbb{Z}^-$)

Rational no. ($\mathbb{Q}, \mathbb{Q}^+, \mathbb{Q}^-$) e.g., $\{\frac{2}{3}, \frac{4}{5}, 0.33, \frac{1}{2}\} \{0.24, 0.2312$

(Repeated value or stops.)

- Irrational numbers: $\{3.454554555\}$ not repeated

$\sqrt{2}, \pi, \frac{22}{7}, e \rightarrow$ exponential ($\sqrt{2}, \sqrt{3}$)

Real no. ($\mathbb{R}, \mathbb{R}^+, \mathbb{R}^-$) Union of rational and irrational no.

- Cardinality of sets: - $|A|$ number of element in set.

$$C = \{x | x \text{ even no. } < 10\} \quad | \quad |C| = 4$$

- Opposite no. e.g. 2 and -2, $-\sqrt{15}$ and $\sqrt{15}$

Reciprocals: $-\frac{1}{5}$ and $-\frac{5}{1}$, $\frac{5}{4}$ and $\frac{4}{5}$

Absolute Value $|x|$

it is distance from 0 either +ive or -ive no.

e.g. $|-5| = 5$ (0 to 5 units) $|\frac{3}{4}| = \frac{3}{4}$

Additive identity $3 + 0 = 3$

Additive inverse $7 + (-7) = 0$

Multiplicative identity $3 \times 1 = 3$

Multiplicative inverse $5 \times \frac{1}{5} = 1$

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* Complex number:-

$$\text{imaginary no. } \left\{ \begin{array}{l} i \Rightarrow \sqrt{-1} \\ i^2 \Rightarrow -1 \end{array} \right\} \rightarrow \text{real numbers}$$

complex numbers

Example:- $x^2 = -4$

$$\sqrt{x^2} = \sqrt{-4}$$

$$x^2 = \sqrt{-1 \times 4}$$

$$\begin{array}{l} x = \sqrt{-1 \times 4} \\ x = 2i \end{array}$$

Complex numbering system: $a + bi$
real imaginary

Example: $\rightarrow \sqrt{-x} = i\sqrt{x}$

$$\rightarrow \sqrt{-25} = i\sqrt{25} = 5i$$

$$\rightarrow \sqrt{-30} = i\sqrt{30}$$

$$\rightarrow \sqrt{-3} \cdot \sqrt{-3} = i\sqrt{3} \cdot i\sqrt{3} = i^2 (\sqrt{3})^2 = (-1)(3) = -3$$

$$\rightarrow \frac{\sqrt{-98}}{\sqrt{49}} = \frac{\sqrt{-1} \sqrt{98}}{\sqrt{49}} = i \frac{\sqrt{98}}{\sqrt{49}} = i\sqrt{2}$$

Example 1: $(4-6i) + (-3+7i)$

$$= 4 - 6i - 3 + 7i = 1 + i$$

2: $(2-4i)(3+5i)$

$$= 6 + 10i - 12i - 20i^2 \Rightarrow 6 - 2i - 20(-1)$$

$$= 6 - 2i + 20 \Rightarrow 26 - 2i$$

$$\because i^2 = -1$$

Conjugate:- $a+bi \rightarrow a-bi$

denoted by $z^*, \bar{z}$

$$1 \quad \frac{3+11i}{-1-2i}$$

$$\frac{3+11i}{-1-2i} \times \frac{-1+2i}{-1+2i} = \frac{(3+11i)(-1+2i)}{(-1)^2 - (2i)^2}$$

$$= \frac{-3 - 25 - 5i}{1+4} = \frac{-28 - 5i}{5} = -\frac{28}{5} - \frac{5i}{5} = -\frac{28}{5} - i$$

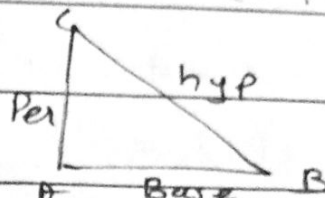
To find absolute value of complex numbers we use pythagoras theorem:-

1. $|-2+5i|$

$$(BC)^2 = (AB)^2 + (AC)^2$$

$$= \sqrt{(-2)^2 + (5)^2}$$

$$= \sqrt{4+25} = \sqrt{29}$$



2. $|-6i|$

$$BC = \sqrt{(AB)^2 + (AC)^2}$$

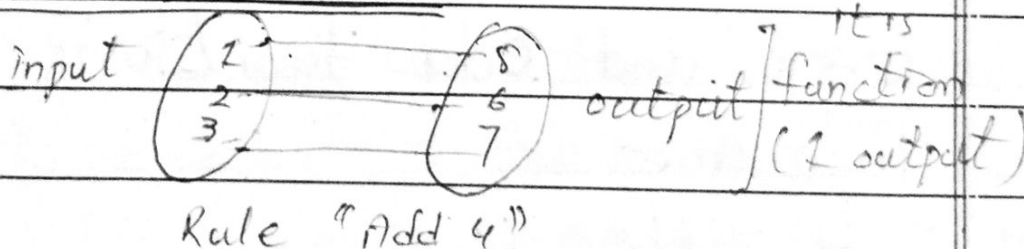
$$= \sqrt{(0)^2 + (-6)^2} = \sqrt{0+36} = \sqrt{36} = 6$$

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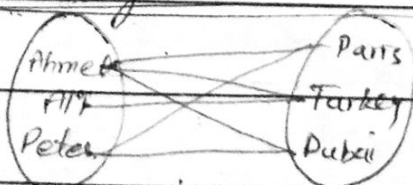
Function and Quadratics

Relation of sets:-

* One to one relation

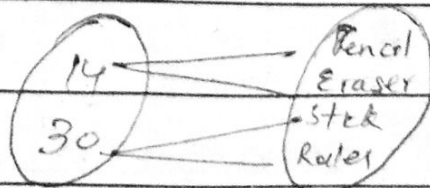


* Many to many relation



Not a function.

* One to many



Not a function

Function Notation:-

$$f(x) = x^2 + 4 \quad \text{eg. } f(2) = 2^2 + 4 = 8$$

$\downarrow$ input output

M T W T F S

Date: _____

(Division and $\sqrt{\quad}$ methods)
range restricted here

Σ

Range

all possible result

Domain

All possible inputs

Example 1) $f(x) = 2x + 1$

Domain $x = \mathbb{R}$

Range $= f(x) = \mathbb{R}$

2) $g(x) = \frac{1}{x-2}$

Domain $x = \mathbb{R} - \{2\}$ Range $g(x) = \mathbb{R}$

3) $h(x) = \sqrt{x-3}$ = sq root of -ive no. are not real no.

$x-3 \geq 0$

$x \geq 3$

Domain: $(x \in \mathbb{R} \mid x \geq 3)$ Range $h(x) \geq 0$

Even and Odd function:-

Theorem:-

a) If $f(-x) = f(x)$ the function is even

b) If $f(-x) = -f(x)$ the f is an odd function

Example 1) $f(x) = x^2$

$f(-x) = (-x)^2$, $f(-x) = x^2$

2) $f(x) = x^3$

(even f)

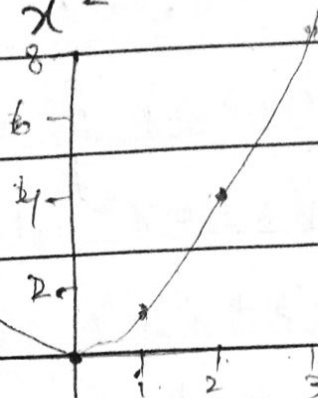
$f(-x) = (-x)^3 \Rightarrow f(-x) = -x^3$ (odd f)

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Graphs of function:-

1- $f(x) = x^2$

parabola shape



$$f'(x) = (x)^2$$

$$f(0) = 0^2 = 0$$

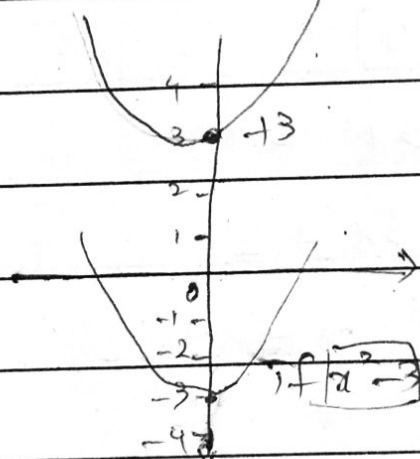
$$f(1) = (1)^2 = 1$$

$$f(2) = (2)^2 = 4$$

$$f(3) = (3)^2 = 9$$

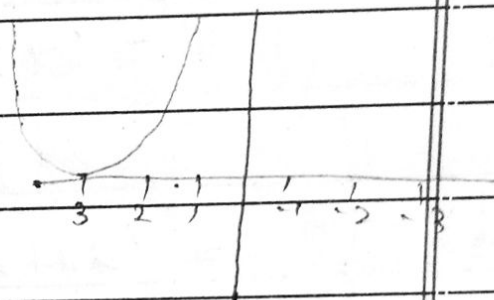
Vertical shift

2. $f(x) = x^2 + 3$

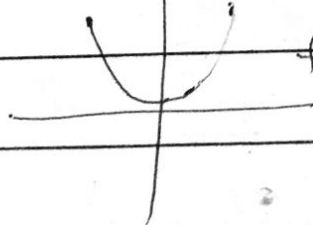


Horizontal shift

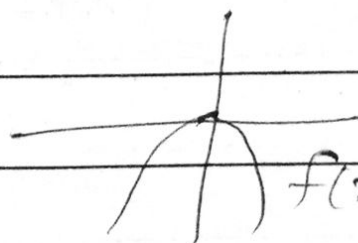
$$f(x) = (x+3)^2$$



Reflection of in graph:-

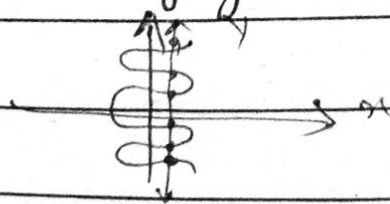


$$f(x) = x^2$$

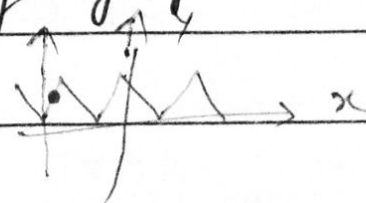


$$f(x) = -x^2$$

Identifying function by graph.



Not function



is function.

(By drawing vertical line, if it cuts at more than one point than it is not a function)

Operation of function:-

Add, subtract, $\times$, $\div$

* We can add, subtract, ^{multi} and divide

function but for division $\frac{f}{g}$ where $g \neq 0$

~~imp~~ Example: $f(x) = 2x + 3$, $g(x) = x^2 - 4$.

$$f(x) + g(x) = 2x + 3 + x^2 - 4$$

$$= x^2 + 2x - 1$$

$$b) f(x) - g(x) = 2x + 3 - (x^2 - 4)$$

$$= 2x + 3 - x^2 + 4 = -x^2 + 2x + 7$$

$$c) f(x) \cdot g(x) = (2x + 3)(x^2 - 4)$$

$$= 2x^3 - 8x + 3x^2 - 12$$

$$= 2x^3 + 3x^2 - 8x - 12$$

$$d) \frac{f(x)}{g(x)} = \frac{(2x + 3)}{(x^2 - 4)}$$

$$x \neq \pm 2$$

$$= \frac{2x + 3}{(x + 2)(x - 2)}$$

$$\because a^2 - b^2 = (a + b)(a - b)$$

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Types of function:-

1- Composite function

2. Inverse function

3. Quadratic function

1 Composite function:- $f \circ g$

Consider 2 function $f(x) = 3x + 2$ and

$$g(x) = x^2$$

$$\therefore f \circ g \Rightarrow f(g(x))$$

$$= f(x^2)$$

$$= 3x^2 + 2$$

$$\therefore g \circ f \Rightarrow g(f(x))$$

$$= g(3x + 2)$$

$$= g(3x + 2)^2$$

Example $f(x) = \sqrt{4-x^2}$, $g(x) = \sqrt{3-x}$

find $f \circ g = f(g(x))$

$$= f(\sqrt{3-x})$$

$$= \sqrt{4 - (\sqrt{3-x})^2}$$

$$= \sqrt{4 - (3-x)}$$

$$= \sqrt{4-3+x} \Rightarrow \sqrt{1+x}$$

2. Inverse function:- f^{-1}

$$f(x) = \frac{5x-2}{3}$$

find f^{-1}

$$f(x) = y$$

① $y = \frac{5x-2}{3}$

② Swap x and y

$$x = \frac{5y-2}{3}$$

③ Solve for y

$$3x = 5y - 2$$

$$5y = 3x + 2$$

$$y = \frac{3x+2}{5}$$

$$y = f^{-1}(x)$$

$$f^{-1}(x) = \frac{3x+2}{5}$$

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3) Quadratic Function:-

$$ax^2 + bx + c$$

constant / coefficients.

graph / Shape $\rightarrow$ parabola

line of symmetry
which cut quadratic function

$$ax^2 + bx + c = 0$$

Factorization method

1	$x^2 = 5x - 6$		
	$x^2 - 5x + 6 = 0$	$x - 3 = 0$	$x - 2 = 0$
	$x^2 - 3x - 2x + 6 = 0$	$x = 3$	$x = 2$
	$x(x-3) - 2(x-3) = 0$		
	$(x-3)(x-2) = 0$		
2.	$2x^2 = 6 = 0$	3. $2x^2 - 3x = 0$	
	$2x^2 = 6 = 0$	$x(2x-3) = 0$	
	$\frac{2x^2}{2} = \frac{6}{2}$	$x = 0$	$2x - 3 = 0$
	$\sqrt{x^2} = \sqrt{3}$		$2x = 3$
	$x = \pm\sqrt{3}$		$x = \frac{3}{2}$

Completing Square Method. (imp)

*	$x^2 + 6x + 3 = 0$		
	$x^2 + 6x = -3$	$\frac{6}{2} = 3$	
	$x^2 + 6x + 9 = -3 + 9$	$(3)^2 = 9$	
	$(x+3)^2 = 6$		
	$\sqrt{(x+3)^2} = \sqrt{6}$		
	$x+3 = \pm\sqrt{6}$		
	$x = -3 \pm\sqrt{6}$		
*	$2x^2 - 16x + 2 = 0$	$\frac{8}{2} = 4^2 = 16$	
	$2(x^2 - 8x + 1) = 0$	$\sqrt{(x-4)^2} = \sqrt{15}$	
	$x^2 - 8x = -1$	$x - 4 = \pm\sqrt{15}$	
	$x^2 - 8x + 16 = -1 + 16$	$x = 4 \pm\sqrt{15}$	

Quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x^2 + 6x + 3 = 0$$

$$x = \frac{-(6) \pm \sqrt{(6)^2 - 4(1)(3)}}{2(1)}$$

$$= \frac{-6 \pm \sqrt{36 - 12}}{2}$$

$$= \frac{-6 \pm \sqrt{24}}{2}$$

$$= \frac{-6 \pm 2\sqrt{6}}{2}$$

$$= -3 \pm \sqrt{6}$$

$$x = -3 \pm \sqrt{6}$$

2	24
2	12
2	6
3	3

LEC 8/ Matrices & Determinants.

Matrix: Rectangular arrangement of numbers in rows and columns.

Order: number of rows & columns

$$\begin{bmatrix} 6 & 2 & -1 \\ -2 & 0 & 5 \end{bmatrix} \text{ order} = 2 \times 3 \quad \left\{ \begin{bmatrix} 2 & 2 \\ 1 & 0 \end{bmatrix} \text{ order} = 2 \times 2 \text{ Square matrix} \right.$$

Entries: Element in matrix.

Operations in matrix.

Addition of Matrix. if same order then we can add.

$$\begin{bmatrix} 5 & -3 \\ -3 & 4 \\ 0 & 7 \end{bmatrix} + \begin{bmatrix} -2 & 1 \\ 3 & 0 \\ 4 & -3 \end{bmatrix} = \begin{bmatrix} 3 & -2 \\ 0 & 4 \\ 4 & 4 \end{bmatrix}$$

Subtraction of matrix. must have same order.

$$\begin{bmatrix} 9 & -2 \\ 5 & 0 \end{bmatrix} - \begin{bmatrix} 4 & 0 \\ 1 & 5 \end{bmatrix}$$

$$\begin{bmatrix} 9-4 & -2-0 \\ 5-1 & 0-5 \end{bmatrix} = \begin{bmatrix} 5 & -2 \\ 4 & -5 \end{bmatrix}$$

Multiplication by a Scalar:

single value,

$$4 \begin{bmatrix} -2 & 0 \\ 4 & -1 \end{bmatrix} = \begin{bmatrix} 4 \times -2 & 4 \times 0 \\ 4 \times 4 & 4 \times -1 \end{bmatrix}$$

$$= \begin{bmatrix} -8 & 0 \\ 16 & -4 \end{bmatrix}$$

Matrix Multiplication:- number of column must be equal to the rows of second matrix.

$$\begin{bmatrix} 1 & 2 \\ 4 & 3 \end{bmatrix} \quad 2 \times 2$$

$$\begin{bmatrix} 4 & 3 & 2 \\ 0 & 1 & 7 \end{bmatrix} \quad 2 \times 3$$

$$= \begin{bmatrix} (1 \times 4) + (2 \times 0) & (1 \times 3) + (2 \times 1) & (1 \times 2) + (2 \times 7) \\ (4 \times 4) + (3 \times 0) & (4 \times 3) + (3 \times 1) & (4 \times 2) + (3 \times 7) \end{bmatrix}$$

$$= \begin{bmatrix} 4+0 & 3+2 & 2+14 \\ 16+0 & 12+3 & 8+21 \end{bmatrix}$$

$$= \begin{bmatrix} 4 & 5 & 16 \\ 16 & 15 & 29 \end{bmatrix}$$

Identity matrix.

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Matrix $\times$ Identity = matrix

Inverse matrix $A^{-1} = \frac{\text{Adj } A}{|A|}$

only fmd for square matrix.

matrix $\times$ inverse = identity matrix

Linear System of Simultaneous Equations.

$$x + y = 6$$

$$2x + y = 9$$

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 1 \end{bmatrix} \quad X = \begin{bmatrix} x \\ y \end{bmatrix} \quad B = \begin{bmatrix} 6 \\ 9 \end{bmatrix}$$

$$AX = B$$

$$X = A^{-1} B$$

$$X = \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 6 \\ 9 \end{bmatrix}$$

$$X = \begin{bmatrix} (-1 \times 6) + (1 \times 9) \\ (2 \times 6) + (-1 \times 9) \end{bmatrix}$$

$$X = \begin{bmatrix} -6 + 9 \\ 12 - 9 \end{bmatrix}$$

$$X = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

$$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix}$$

$$= (1 \times 1) - (1 \times 2)$$

$$= 1 - 2$$

$$= -1 \neq 0$$

$$A^{-1} = \frac{1}{|A|} \begin{bmatrix} 1 & -1 \\ -2 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{1}{-1} \begin{bmatrix} 1 & -1 \\ -2 & 1 \end{bmatrix}$$

$$A^{-1} = \begin{bmatrix} -1 & 1 \\ 2 & -1 \end{bmatrix}$$

Properties of Matrix Multiplication:

• $AB \neq BA$ do not commute.

• $A(BC) = (AB)C$ Associative property

• $A(B+C) = AB+AC$ Left distributive property

$(B+C)A = BA+CA$ right distributive property

$A=B$ then $AC=BC$

$k(AB) = (kA)B = A(kB)$

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Right multiplicative property

Echelon form.

$$\begin{bmatrix} 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 & 2 & 3 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Properties.

- 1- non-zero row start from 1.
- 2- every lower row (1) is to the right of above
- 3- If there are zero row they are at the end

Reduce Echelon form.

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{bmatrix} 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

non zero values are (1) and all the rest values (1) are zero.

Solve the system of equation.

$$x + 2y + z = 1$$

$$2x + 2y = 2$$

$$3x + 5y + 4z = 1$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 2 & 0 \\ 3 & 5 & 4 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 & : & 1 \\ 2 & 2 & 0 & : & 2 \\ 3 & 5 & 4 & : & 1 \end{bmatrix} \quad R_2 - 2R_1$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 3 \\ 0 & -2 & -2 & 0 \\ 3 & 5 & 4 & 1 \end{array} \right]$$

$$R_3 - 3R_1$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 3 \\ 0 & -2 & -2 & 0 \\ 0 & -1 & 1 & -2 \end{array} \right] \xrightarrow{-\frac{1}{2}R_2}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 3 \\ 0 & 1 & 1 & 0 \\ 0 & -1 & 1 & -2 \end{array} \right]$$

$$R_3 - 2R_2$$

$$R_3 + R_2$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 2 & -2 \end{array} \right]$$

$$\xrightarrow{\frac{1}{2}R_3}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1 \end{array} \right]$$

$A \times B$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 1 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & -1 \end{array} \right] \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

$$z = -1$$

$$y + z = 0$$

$$y + (-1) = 0$$

$$\boxed{y = 1}$$

$$x + z = 1$$

$$x + (-1) = 1$$

$$x + 1 = 1$$

$$x = 0$$

$$x = 0$$

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Inverse using row operation.

$$A = \begin{bmatrix} 2 & -2 & 2 \\ 0 & 2 & -1 \\ 2 & 3 & 0 \end{bmatrix} \quad \text{Find inverse.}$$

$$A = \begin{bmatrix} 2 & -2 & 2 \\ 0 & 2 & -1 \\ 2 & 3 & 0 \end{bmatrix} \quad I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$A|I = \left[\begin{array}{ccc|ccc} 2 & -2 & 2 & 1 & 0 & 0 \\ 0 & 2 & -1 & 0 & 1 & 0 \\ 2 & 3 & 0 & 0 & 0 & 1 \end{array} \right] \quad \frac{1}{2} R_1$$

$$= \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & \frac{1}{2} & 0 & 0 \\ 0 & 2 & -1 & 0 & 1 & 0 \\ 2 & 3 & 0 & 1 & 0 & 1 \end{array} \right] \quad R_3 - 2R_1$$

$$= \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & \frac{1}{2} & 0 & 0 \\ 0 & 2 & -1 & \frac{1}{2} & 0 & 0 \\ 0 & 5 & -2 & 0 & 0 & 1 \end{array} \right] \quad \frac{1}{2} R_2$$

$$= \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & \frac{1}{2} & 0 & 0 \\ 0 & 1 & -\frac{1}{2} & 0 & \frac{1}{2} & 0 \\ 0 & 5 & -2 & 0 & 0 & 1 \end{array} \right]$$

$$= \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & \frac{3}{2} & 3 & -1 \\ 0 & 1 & 0 & 1 & -2 & 1 \\ 0 & 0 & 1 & -2 & -5 & 2 \end{array} \right]$$

A^{-1}

LEC: 11

Determinants:- process of attaching a real no. to every square matrix.

Example:- $\begin{bmatrix} 2 & -3 \\ 5 & 1 \end{bmatrix}$

$$\begin{aligned} |D| &= (2 \times 1) - (5 \times -3) \\ &= 2 - (-15) \\ &= 17 \text{ Ans} \end{aligned}$$

$$\begin{bmatrix} 2 & 0 & 3 \\ 1 & -2 & 5 \\ -1 & 4 & 2 \end{bmatrix}$$

$$= 2 \begin{bmatrix} -2 & 5 \\ 4 & 2 \end{bmatrix} - 0 \begin{bmatrix} 1 & 5 \\ -1 & 2 \end{bmatrix} + 3 \begin{bmatrix} 1 & -2 \\ -1 & 4 \end{bmatrix}$$

$$= 2(-4 - 20) - 0 + 3(4 - 2)$$

$$= 2(-24) + 3(2) = -48 + 6 = \boxed{-42}$$

Cofactor:- The cofactor of $a_{ij} = (-1)^{i+j}$ (Minor of a_{ij})

Example: $\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

$i + j = \text{row} + \text{column}$

$$\text{Cofactor of } a_{11} = C_1 = (-1)^{1+1} \begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix}$$

$$= + \begin{bmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{bmatrix}$$

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Example 1 from book

$$\begin{bmatrix} 2 & -2 & 0 \\ -3 & 1 & 2 \\ 1 & -3 & -1 \end{bmatrix}$$

$$= 2 \left[(-1)^{1+1} \begin{vmatrix} 1 & 2 \\ -3 & -1 \end{vmatrix} \right] + (-2) \left[(-1)^{1+2} \begin{vmatrix} -3 & 2 \\ 1 & -1 \end{vmatrix} \right] + 0 \left[(-1)^{1+3} \begin{vmatrix} -3 & 1 \\ 1 & -3 \end{vmatrix} \right]$$

$$= 2(1)(-1-6) + 2(3-2) + 0$$

$$= 2(5) + 2(1) = 10 + 2 = 12$$

Inverse using Determinant:-

Transpose A^T :- obtain by inverting rows of A with column. e-g.

$$A = \begin{bmatrix} 8 & 1 & 4 \\ 0 & 2 & -5 \\ 10 & 3 & 6 \end{bmatrix} \quad A^T = \begin{bmatrix} 8 & 0 & 10 \\ 1 & 2 & 3 \\ 4 & -5 & 6 \end{bmatrix}$$

adj is transport of matrix

Inverse Formula

$$A^{-1} = \frac{1}{|A|} \text{Adj } A$$

Example:- $A = \begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$

$$|A| = (4 - 6) = -2 \quad A^T = \begin{vmatrix} 4 & -2 \\ -3 & 1 \end{vmatrix}$$

$$A^{-1} = \frac{1}{-2} \begin{vmatrix} 4 & -2 \\ -3 & 1 \end{vmatrix}$$

$$= \begin{bmatrix} -\frac{1}{2} \times 4 & -\frac{1}{2} \times (-2) \\ -\frac{1}{2} \times (-3) & -\frac{1}{2} \times 1 \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ \frac{3}{2} & -\frac{1}{2} \end{bmatrix}$$

$$A = \begin{bmatrix} -2 & 1 & 3 \\ 0 & -1 & 1 \\ 1 & 2 & 0 \end{bmatrix} \quad \text{Find } A^{-1}$$

$$|A| = -2 \begin{vmatrix} -1 & 1 \\ 2 & 0 \end{vmatrix} - 0 \begin{vmatrix} 1 & 3 \\ 2 & 0 \end{vmatrix} + 1 \begin{vmatrix} 1 & 3 \\ -1 & 1 \end{vmatrix}$$

$$= -2(0-2) - 0 + 1(1+3)$$

$$= 4 + 4 = 8$$

We need to find cofactors first

$$C_{11} = (-1)^{1+1} \begin{vmatrix} -1 & 1 \\ 2 & 0 \end{vmatrix} = 1(0-2) = \boxed{-2}$$

$$C_{12} = (-1)^{1+2} \begin{vmatrix} 0 & 1 \\ 1 & 0 \end{vmatrix} = -1(0-1) = \boxed{1}$$

$$C_{13} = (-1)^{1+3} \begin{vmatrix} 0 & -1 \\ 1 & 2 \end{vmatrix} = 1(0+1) = \boxed{1}$$

$$C_{21} = (-1)^{2+1} \begin{vmatrix} 1 & 3 \\ 2 & 0 \end{vmatrix} = -1(0-6) = \boxed{6}$$

$$C_{22} = (-1)^{2+2} \begin{vmatrix} -2 & 3 \\ 1 & 0 \end{vmatrix} = 1(0-3) = \boxed{-3}$$

$$C_{23} = (-1)^{2+3} \begin{vmatrix} -2 & 1 \\ 1 & 2 \end{vmatrix} = -1(-4-1) = \boxed{5}$$

$$C_{31} = (-1)^{3+1} \begin{vmatrix} 1 & 3 \\ -1 & 1 \end{vmatrix} = 1(1+3) = \boxed{4}$$

$$C_{32} = (-1)^{3+2} \begin{vmatrix} -2 & 3 \\ 0 & 1 \end{vmatrix} = -1(-2-0) = \boxed{2}$$

$$C_{33} = (-1)^{3+3} \begin{vmatrix} -2 & 1 \\ 0 & -1 \end{vmatrix} = 1(2-0) = \boxed{2}$$

$$\text{Adj } A = C^T = \begin{bmatrix} -2 & 6 & 4 \\ 1 & -3 & 2 \\ 1 & 5 & 2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{8} \begin{bmatrix} -2 & 6 & 4 \\ 1 & -3 & 2 \\ 1 & 5 & 2 \end{bmatrix}$$

Cramer's Rule.

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{\det A} \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix}$$

$$x_1 = \frac{\begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix}}{\det A} \quad x_1 = \frac{\Delta_1}{\Delta}$$

$$x_2 = \frac{\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}}{\det A} = x_2 = \frac{\Delta_2}{\Delta}$$

Example Solve eq by Cramer's rule

$$3x - 5y = 2$$

$$-4x + 3y = -1$$

$$A = \begin{bmatrix} 3 & -5 \\ -4 & 3 \end{bmatrix} \quad \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$$

$$|A| = 9 + 20 = 11$$

$$x = \frac{|x_1|}{|A|} = \frac{\begin{vmatrix} 2 & -5 \\ -1 & 3 \end{vmatrix}}{11} = \frac{6 - 5}{11} = 1$$

$$y = \frac{|x_2|}{|A|} = \frac{\begin{vmatrix} 3 & 2 \\ -4 & -1 \end{vmatrix}}{11} = \frac{-3 + 8}{11} = 5$$

$$x = \frac{|x_1|}{|A|} = \frac{1}{11}$$

$$y = \frac{|x_2|}{|A|} = \frac{-5}{11}$$

LEC : 13

Sequence & Series

Sequences

- 1, 4, 9, 16 — — — — — $1^2, 2^2, 3^2$
- $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}$ — — — — — $(\frac{n}{n+1})$
- 99, 97, 95, 93 — — — — — (subtract 2 from previous)
- 1, 1-1, 1-21, 1-331, 1-4641, — — — — — (obtain by "x" each by (1.1))
- $\begin{matrix} +12 & +4 & +6 & +8 & +10 & +12 & +14 \\ 2 & 4 & 8 & 14 & 22 & 32 & 44 & 58 \end{matrix}$ (add 2 in each pattern)
 $\begin{matrix} 2+2 & 4+4 & 8+6 & 14+8 & 22+10 & 32+12 & 44+14 \end{matrix}$
- 1, 1, 2, 3, 5, 8, 13 — — — — — (Add previous 2 terms)

Rows of no. of these type are called sequence & separated no. called term

Terms showed by $a_1, a_2, a_3, a_4, \dots, a_n$

Types :- Arithmetic Sequence -

Geometric Sequence

Arithmetic Sequence - each term is adding constant common difference

$$a_1 = a$$

$$a_2 = a_1 + d$$

$$a_3 = a_2 + d$$

$$a_n = a_{n-1} + d$$

$$d = a_2 - a_1 = 5 - 2 = 3$$

Formula = $a_n = a + (n-1)d$ if 50 terms = ?

$$a_{50} = 2 + (49)3$$

$$a_{50} = 149 \text{ ans}$$

Example: 72 is term of sequence -5, 2, 9,
we need to find "n" term of number
 $a_n = 72$ $d = 7$

$$a_n = a + (n-1)d$$

$$72 = -5 + (n-1)7$$

$$72 = -5 + 7n - 7$$

$$72 + 5 - 7 = 7n$$

$$84 = 7n$$

$$\frac{84}{7} = n$$

$$n = 12$$

LEC : 14

Arithmetic Series :- (+)

Sum of terms in arithmetic
sequence. 1, 2, 3, 4, 5 ar. sequence

1 + 2 + 3 + 4 + 5 — ar. series

$$\text{Formula} = S_n = \frac{n}{2} (a + a_n)$$

$$\text{if } a_n = a + (n-1)d.$$

Exampler- Find sum of 10th term of a-series with $a=6$, $a_{10}=151$

$$S_n = \frac{n}{2} (a + a_n)$$

$$a_{10} = a + 9d$$

$$S_1 = 6 + 9d$$

$$S_1 - 6 = 9d$$

$$\frac{151-6}{9} = d$$

$$d = 5$$

$$S_{10} = 5 (87)$$

$$S_{10} = 285$$

Find sum of first 50 term of a series with $a = 24$, $d = -4$.

$$S_n = \frac{n}{2} (a + a_n)$$

$$a_n = a + (n-1)d$$

$$a_{50} = a + 49d$$

$$a_{50} = 24 + 49(-4)$$

$$= 24 - 196$$

$$a_{50} = -172$$

$$S_{50} = \frac{50}{2} (24 + (a + (n-1)d))$$

$$S_{50} = 25 (24 + (24 + 49(-4)))$$

$$S_{50} = 25 (24 + 24 - 196)$$

$$S_{50} = 25 (-148)$$

$$S_{50} = -3500 \text{ Ans}$$

Geometric Sequence:- "x"

Obtain by "x" previous term by constant value called common ratio.

Example:- Find 5th term of G-S with $a = -3$, C.R = -8 .

$$-3, -15, -75, -375, -1875$$

$$\text{Common ratio} = a_2 \div a_1 \\ = a$$

G.S Formula:-

$$a_n = a \cdot r^{n-1}$$

Exampler Find 10th term of G.S
with $a = 2000$, common ratio $= r = \frac{1}{2}$

$$a_n = a \cdot r^{n-1}$$

$$a_{10} = (2000) \left(\frac{1}{2} \right)^{10-1}$$

$$a_{10} = 2000 \left(\frac{1}{2} \right)^9$$

$$a_{10} = 2000 \left(\frac{1}{512} \right)$$

$$a_{10} = \frac{125}{32}$$

Find next two sum in sequence

$$-64, -16, -4, _, _$$

$$\text{Common ratio} = r = -16 \div -64 \\ = 1/4$$

$$-4 \times 1/4 = -1$$

$$-1 \times 1/4 = -1/4$$

$$\text{So, } -64, -16, -4, -1, -1/4$$

Geometric Series:- Sum of
geometric Series

Formula

$$S_n = \frac{a(1-r^n)}{1-r}$$

Exp 1 - Find S_{12} when in G-Series
 $a = 400$ and $r = 8$

$$S_n = \frac{a(1-r^n)}{1-r}$$

$$S_{12} = \frac{400(1-8^{12})}{1-8}$$

$$= \frac{400(1-68719)}{7}$$

$$= \frac{-27487}{7}$$

$$S_{12} = 3926787 \times 10^{12}$$

LEC : 15

Sequences and Series.

Convergence: - If sequence of no. approaches to finite no. we say that sequence is convergent



Diverger

Does not converge to finite no.



Infinite Sum of Geometric Series:

$$S_{\infty} = \frac{a}{1-r}$$

Expi: - Express the recurring decimal

$0.296296296\ldots$ as a fraction.

$$0.296 + 0.000296 + 0.000000296 + \ldots$$

$$a_1 = 0.296$$

$$r = \frac{0.000296}{0.296} = 0.001$$

$$S_{\infty} = \frac{a}{1-r}$$

$$S_{\infty} = \frac{0.296}{1 - 0.001}$$

$$= \frac{0.296}{0.999} = \frac{296}{999} \quad \text{Ans.}$$

Evaluate $\sum_{n=1}^{\infty} 35 \left(-\frac{1}{4}\right)^{n-1}$

$$35 \left(-\frac{1}{4}\right)^0 + 35 \left(-\frac{1}{4}\right)^1 + 35 \left(-\frac{1}{4}\right)^2 + 35 \left(-\frac{1}{4}\right)^3 \dots$$

$$a = 35$$

$$r = \left(-\frac{1}{4}\right)$$

$$a_1 = a_2 = a_3 = \dots = a_n$$

$$r = \frac{a_2}{a_1} = \frac{35 \left(-\frac{1}{4}\right)}{35}$$

$$S_{\infty} = \frac{a}{1-r}$$

$$= \frac{35}{1 + \frac{1}{4}}$$

$$= \frac{35}{\frac{5}{4}} = \frac{35 \times 4}{5} = \frac{140}{5} = 28$$

$$\boxed{S_{\infty} = 28 \text{ Ans}}$$

REVISION

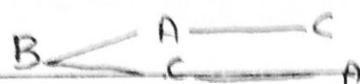
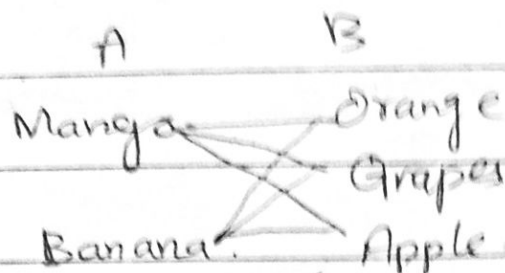
LEC = 19.

Permutation:-

- It is a technique for counting.
- order is important eg lock.

Multiplication Principle:-

$$\textcircled{5} \times \textcircled{3} = 6$$



$$3 \times 2 \times 1 = 6$$

Rule:-

$$n \times (n-1) \times (n-2) \times (n-3) \dots 3 \times 2 \times 1 = n!$$

Total letter $\rightarrow n! = n$ -factorial

Different arrangement of the objects are called permutation.

Example:- A, B, C, D, E, F, G and

you want to make 4 letter words

$$7 \times 6 \times 5 \times 4 = 840$$

$$\frac{(7)(6)(5)(4)(3)(2)(1)}{(3)(3)(1)} = \frac{7!}{3!} = \frac{7!}{(7-4)!}$$

$${}^n P_r = \frac{n!}{(n-r)!} = {}_n P_r$$

Q. Example - 8 runner hoping to take part in race, but we have only 6 lanes. How many ways can some of 8 members arrange

$${}_n P_r = \frac{n!}{(n-r)!}$$

$${}_8 P_6 = \frac{8!}{(8-6)!}$$

$$= (8) \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1$$

$$= 2016$$

$${}_4 P_4 = \frac{4!}{(4-4)!} = \frac{4!}{0!} = 4! \because 0! = 1$$

8 Find no. of distinct permutations of letters word Mississippi

Total letters = 11 No. of Is = 4

No. of Ss = 4 No. of Ps = 2

$$\frac{n!}{P! Q! R!} = \frac{11!}{4! 4! 2!} = \frac{11 \cdot 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{4 \cdot 3 \cdot 2 \cdot 1 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \cdot 2 \cdot 1}$$

$$= 34650$$

2. Five people A, B, C, D, E are arranged randomly in line. Find possible permutations when A & B next to each other

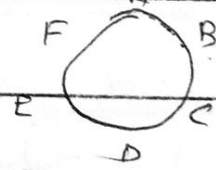
i) $AB, C, D, E = 4! \quad 4! + 4! = 48$
 $BA, C, D, E = 4! \quad 2(4!) = 48$

ii) $A, B, C, D, E, S! = 120$
 $= 120 - 48 = 72$

Rule $r! (n-r+1)!$

$2! (5-4+1)! \quad 2! (5-4+1)!$
 $2(2)! = 4! = 48 \quad 2!(2)! = 4! =$

Q. How many ways are there in to sit 6 people around a circular table?



Fix one ${}^5P_5 = 5! = 120$ ways.

${}^6P_6 = 6! = 720$ ways $\frac{720}{6} = 120$

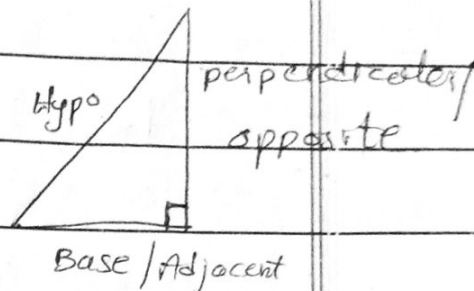
Rule $\frac{n!}{n} = (n-1)!$

LEC 25:- Trigonometry.

$$\cos \theta = \frac{\text{adjacent}}{\text{Hypo}} = \frac{B}{H}$$

$$\sin \theta = \frac{\text{Opposite}}{\text{Hypo}} = \frac{P}{H}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{\sin \theta}{\cos \theta} = \frac{y}{x}$$



- anti-clock wise $\rightarrow$ +ve motion
- clock-wise $\rightarrow$ -ve motion



Defining cosine function:- (x)

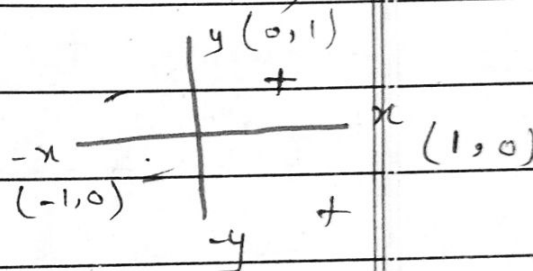
$$\cos 0^\circ = 1$$

$$\cos 90^\circ = 0$$

$$\cos 180^\circ = -1$$

$$\cos 270^\circ = 0$$

$$\cos 360^\circ = 1$$



Properties of cos function:-

- * Cos function has period of 360° and repeat it self after each revolution.

$$1- \cos(-\theta) = \cos \theta$$

$$2- \cos(180^\circ - \theta) = -\cos \theta$$

$$3- \cos(\theta - 180^\circ) = -\cos \theta$$

$\cos \theta$ is positive in 1st and 4th quadrant.

1st and 4th

Defining sin function:-

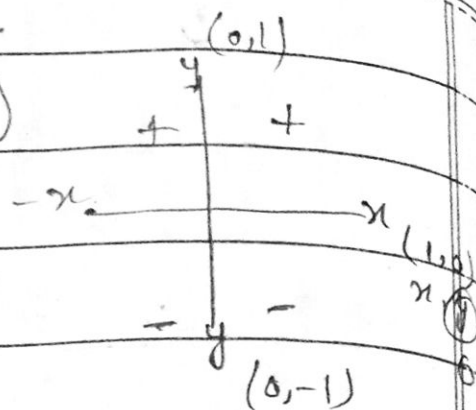
$$\sin 0^\circ = 0$$

$$\sin 90^\circ = 1$$

$$\sin 180^\circ = 0$$

$$\sin 270^\circ = -1$$

$$\sin 360^\circ = 0$$



Properties:-

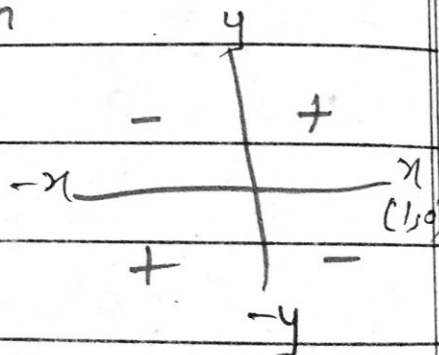
$$\sin(-\theta) = -\sin \theta, \quad \sin(\theta - 180) = -\sin \theta$$

$$\sin(180 - \theta) = \sin \theta \quad \bullet \text{ positive in } \boxed{1 \ \& \ 2}$$

negative in $(3 \ \& \ 4)$

Defining tan function

$$\tan \theta = \frac{y}{x}$$



$$\tan 0^\circ = 0 \leftarrow \frac{0}{1}$$

$$\tan 90^\circ = \text{undefined} \leftarrow \frac{1}{0}$$

$$\tan 180^\circ = 0$$

$$\tan 270^\circ = \text{undefined}$$

$$\tan 360^\circ = 0$$

Properties:-

$$\tan(-\theta) = -\tan \theta, \quad \tan(\theta - 180) = \tan \theta$$

$$\tan(180 - \theta) = -\tan \theta \quad \bullet \text{ positive in } 1 \text{ and } 3$$

negative in $2 \ \& \ 4$.

Radians.

π radian = 180 degrees.

Example $135^\circ \rightarrow$ radian

$$= \frac{135^\circ \times \pi}{180} = \frac{3}{4} \pi$$

Exp 2. 22.5°

$$\frac{22.5^\circ \times \pi}{180} = \frac{5}{8} \pi$$

$$= \frac{5}{8} \pi = \frac{5}{8} \pi$$

LEC 26

Function Periodic Properties odd (even).

Cos $0 \pm 360 \left. \vphantom{\begin{matrix} 0 \pm 360 \\ 0 \pm 360 \end{matrix}} \right\} 2\pi$ even $\{ \cos(-\theta) = \cos \theta$

Sm $0 \pm 360 \left. \vphantom{\begin{matrix} 0 \pm 360 \\ 0 \pm 360 \end{matrix}} \right\} 2\pi$ $\{ \sin(-\theta) = -\sin \theta$

Tan $0 \pm 360 \left. \vphantom{\begin{matrix} 0 \pm 360 \\ 0 \pm 360 \end{matrix}} \right\} \pi$ odd $\{ \tan(-\theta) = -\tan \theta$

Function Transational properties

Cos $\cos(\theta - 180) = -\cos \theta$

$\cos(180 - \theta) = -\cos \theta$

Sm $\{ \sin(\theta - 180) = -\sin \theta \}$

$\sin(180 - \theta) = \sin \theta$

Tan $\tan(\theta - 180) = \tan \theta$

$\{ \tan(180 - \theta) = -\tan \theta \}$

θ degrees	θ radians	$\sin \theta$	$\cos \theta$	$\tan \theta$
30°	$\frac{\pi}{6}$	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$1/\sqrt{3}$
45°	$\frac{\pi}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{2}}{2}$	1
60°	$\frac{\pi}{3}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$
90°	$\frac{\pi}{2}$	1	0	∞

Example:-

$$\cos(135) = \cos(180 - 45) = -\cos 45 = -\frac{\sqrt{2}}{2}$$

$$\sin(120) = \sin(180 - 60) = \sin 60 = \frac{\sqrt{3}}{2}$$

$$\tan 495 = \tan(495 - 180) = \tan 315$$

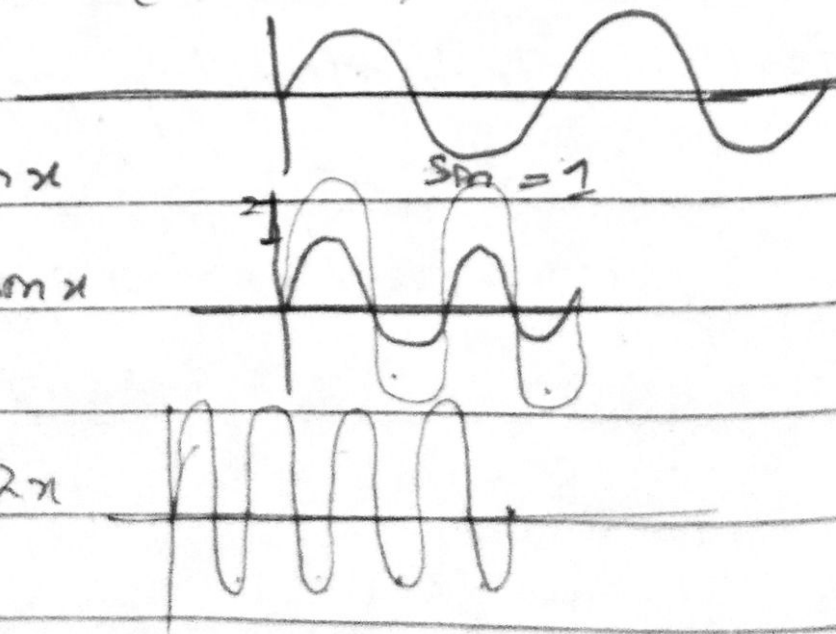
$$\tan 315 = \tan(315 - 180) = \tan 135$$

$$\tan 135 = \tan(180 - 45) = -\tan 45 = -1$$

$$y = \sin x$$

$$y = 2 \sin x$$

$$y = 2 \sin 2x$$



Inverse function:-

$$y = \sin x$$

$$y = \arcsin x$$

$\sin, \cos, \tan$

$$y = \sin^{-1} x$$

not one to one function to find inverse
we restrict domain.

Example $\sin^{-1} \frac{1}{2} = \frac{\pi}{6} = 30^\circ$ Sm inverse

$\frac{\pi}{6}$ is angle where $\sin$ is $\frac{1}{2}$.

$$\sin^{-1} \frac{\sqrt{3}}{2} = \frac{\pi}{3} = 60^\circ$$

Cos inverse

$$\cos^{-1} \frac{1}{2} = \frac{\pi}{3} = 60^\circ$$

$$\cos^{-1} \left(-\frac{\sqrt{3}}{2} \right) = \frac{5\pi}{6} = 150^\circ$$

Tan inverse.

Domain of $y = \arctan = (-\infty, \infty)$

Range of $y = \arctan = \left(-\frac{\pi}{2}, \frac{\pi}{2} \right)$

$$\tan \theta = \frac{y}{x}$$

$$\tan^{-1} \frac{\sqrt{3}}{3}$$

$$\frac{\frac{\sqrt{3}}{3} \times \sqrt{3}}{\sqrt{3}} = \frac{1}{\sqrt{3}} = \frac{\pi}{6} = 30^\circ$$

$$\begin{array}{c|c} S & A \\ \hline T & C \end{array}$$

Example $\cos x = 0.5$ $0^\circ \leq x \leq 360^\circ$

Step 1 =
 $\cos$ positive in 1st & 4th quadrant

Step 2
 $x = \cos^{-1}(0.5) = 60^\circ$

Step 3. $\cos(-\theta) = \cos \theta$
 $\cos(-60) = \cos 60^\circ = 0.5$

Step 4: $\cos(\theta \pm 360) = \cos \theta$
 $\cos(-60 + 360) = \cos 300$
 $x = (60, 300)$

$$\cos 3\theta = -\frac{1}{2} \quad -180^\circ \leq \theta \leq 180^\circ$$

Adjust range by "x" 3..

$$-540 \leq \theta \leq 540^\circ$$

Step 1 = $\cos$ neg in 2 and 3

Step 2 = $\cos^{-1}\left(-\frac{1}{2}\right) = -60^\circ$

Step 3 = $\cos(-\theta) = \cos \theta$
 $\cos -(-60^\circ) = \cos(-60^\circ)$

Step 4 = $\cos(\theta \pm 360) = \cos \theta$

-240
-180
-60
180
-180
-60
180
240

$$\sin x = -0.5 ; -360^\circ < x < 360^\circ$$

Step 1 = sin is neg in 3 & 4

$$\text{Step 2} = \sin^{-1}(-0.5) = -30^\circ$$

$$\text{Step 3} = x = \sin(-\theta) = -\sin \theta$$

$$\sin(180 - \theta) = \sin \theta$$

$$\sin(180 - (-30)) = \sin(-30) = \sin 210^\circ$$

$$\text{Step 4} = \sin(\theta \pm 360) = \sin \theta$$

$$(-30 + 360) = 330$$

$$(30 + 360) = 390$$

$$(210 + 360) = 570$$

$$(210 - 360) = -150$$

$$x = -30^\circ, 210^\circ, 330^\circ, -150^\circ$$

Q28 Tangent equations

$$3 + 5 \tan 2x = 0 ; 0^\circ \leq x \leq 360^\circ$$

$$5 \tan 2x = -3$$

$$\tan 2x = -\frac{3}{5} ; 0^\circ \leq 2x \leq 720^\circ$$

Step 1 = Tan neg in 2 & 4 quadrant.

~~tan~~

$$\text{Step 2} = \tan^{-1}\left(-\frac{3}{5}\right) = -30.96^\circ$$

$$\text{Step 3} = \tan(\theta \pm 180)$$

$$2x = (-30.96 + 180) = 149.04$$

$$= (149.04 + 180) = 329.04$$

$$= (329.04 + 180) = 509.04$$

$$= (509.04 + 180) = 689.04$$

74.5

164.5

254.5

344.5

احمد بیبر پروانکشا

$$2 \sin(4x + 90^\circ) - 1 = 0$$

$$0 < x < 90^\circ$$

$$\sin 4x + 90^\circ = \frac{1}{2} ; 0 < 4x < 360$$

$$90^\circ < 4x + 90^\circ < 450^\circ$$

$$\text{Let } 4x + 90^\circ = \theta ; 90^\circ < \theta < 450^\circ$$

$$\sin \theta = \frac{1}{2}$$

Step 1

$$\text{Basic angle } \theta = \sin^{-1} = \underline{30^\circ} \times$$

Symmetry Property:-

$$\sin(180 - \theta) = 180 - 30 = \underline{150} \checkmark$$

Periodic property

$$\sin(\theta \pm 360)$$

$$(30 + 360) = 390 \checkmark$$

$$(30 - 360) = 330 \checkmark$$

$$\begin{array}{r} 360 \\ 150 \\ \hline 210 \end{array}$$

$$(150 + 360) = 510 \times$$

$$(150 - 360) = 210 \checkmark$$

$$\cancel{x} \theta, 150, 210, 330, 390$$

$$4x + 90 = 150 - 90, 210 - 90, 330 - 90, 390 - 90$$

$$x = \frac{60}{4}, \frac{120}{4}, \frac{240}{4}, \frac{300}{4}$$

$$x = 15, 75, 30, 60,$$

$$\sin(19.5^\circ) = \frac{1}{3} \quad \text{Solve } 6\sin^2 x + \sin x - 1 = 0$$

$$; 0 \leq x \leq 360^\circ$$

$$6\sin^2 x + \sin x - 1 = 0$$

$$\text{put } \sin x = y$$

$$6y^2 + y - 1 = 0$$

$$6y^2 + 3y - 2y - 1 = 0$$

$$3y(2y+1) - 1(2y+1) = 0$$

$$(2y+1)(3y-1) = 0$$

$$3y - 1 = 0$$

$$y = \frac{1}{3}$$

$$\sin x = \frac{1}{3}$$

$$\sin^{-1} \frac{1}{3} = 19.5^\circ \checkmark$$

$$1. \sin(180^\circ - \theta) = (180 - 19.5)$$

$$= 160^\circ \checkmark$$

$$2. \sin(19.5 + 360) = 379.5^\circ$$

$$(160 + 360) = 520^\circ \times$$

$$2y + 1 = 0$$

$$y = -\frac{1}{2}$$

$$\sin x = -\frac{1}{2}$$

$$\sin^{-1}\left(-\frac{1}{2}\right) = (-30^\circ) \times$$

$$2. \sin(180 - (-30)) = 210^\circ \checkmark$$

$$3. \sin(-30 + 360)$$

$$= 330^\circ \checkmark$$

$$\sin(210 + 360)$$

$$= 570^\circ \times$$

$$x = 19.5, 160, 210, 330$$

Identities

$$f(x) \equiv g(x)$$

identity

$$f(x) = g(x)$$

equal.

Reciprocal identity:-

$$\sin \theta = \frac{1}{\csc \theta}$$

$$\csc \theta = \frac{1}{\sin \theta}$$

$$\cos \theta = \frac{1}{\sec \theta}$$

$$\sec \theta = \frac{1}{\cos \theta}$$

$$\tan \theta = \frac{1}{\cot \theta}$$

$$\cot \theta = \frac{1}{\tan \theta}$$

Quotient identity

$$\tan \theta = \frac{\sin \theta}{\cos \theta}, \cot \theta = \frac{\cos \theta}{\sin \theta}$$

Periodic properties:-

$$\sin(\theta + 2\pi) = \sin \theta, \csc(\theta + 2\pi) = \csc \theta$$

$$\cos(\theta + 2\pi) = \cos \theta, \sec(\theta + 2\pi) = \sec \theta$$

$$\tan(\theta + \pi) = \tan \theta, \cot(\theta + \pi) = \cot \theta$$

Pythagoras identities:-

$$\sin^2 \theta + \cos^2 \theta = 1$$

$$x^2 + y^2 = 1$$

$$\tan^2 \theta + 1 = \sec^2 \theta$$

$$1 + \cot^2 \theta = \csc^2 \theta$$

Example:- $\sin \theta \operatorname{cosec} \theta - \cos^2 \theta = \sin^2 \theta$

$$\text{L.H.S} = \sin \theta \operatorname{cosec} \theta - \cos^2 \theta$$

$$= \cancel{\sin \theta} \times \frac{1}{\cancel{\sin \theta}} - \cos^2 \theta$$

$$= 1 - \cos^2 \theta$$

$$= \sin^2 \theta$$

LEC = 29.

Example:- $\frac{\cos \theta + 1}{\cos \theta - 1} = \frac{1 + \sec \theta}{1 - \sec \theta}$

$$\text{R.H.S} = \frac{1 + \sec \theta}{1 - \sec \theta}$$

$$= \frac{1 + \frac{1}{\cos \theta}}{1 - \frac{1}{\cos \theta}}$$

$$= \frac{\frac{\cos \theta + 1}{\cos \theta}}{\frac{\cos \theta - 1}{\cos \theta}}$$

$$= \frac{\cos \theta + 1}{\cos \theta - 1}$$

$$= \frac{\cos \theta + 1}{\cos \theta - 1} \text{ Ans.}$$

Example:- $\frac{(1 + \sin \theta)^2}{1 - \sin \theta} - \frac{(1 - \sin \theta)^2}{1 + \sin \theta} = 4 \tan \theta \sec \theta$

$$\text{R.H.S} = (1 + \sin^2 \theta + 2 \sin \theta) - (1 + \sin^2 \theta - 2 \sin \theta)$$

$$(1)^2 (\sin \theta)^2$$

$$= \frac{4 \sin \theta}{1 - \sin^2 \theta} \Rightarrow \frac{4 \sin \theta}{\cos^2 \theta} \Rightarrow \frac{4 \sin \theta}{(\cos \theta)(\cos \theta)} = 4 \tan \theta \sec \theta$$

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$$\frac{\tan t - \cot t}{\sin t \cos t} = \sec^2 t - \csc^2 t$$

$$\text{L.H.S} = \frac{\tan t - \cot t}{\sin t \cos t}$$

$$= \frac{\frac{\sin t}{\cos t} - \frac{\cos t}{\sin t}}{\sin t \cos t}$$

$$= \frac{\sin^2 t - \cos^2 t}{\cos t \sin t}$$

$$= \frac{\sin^2 t - \cos^2 t}{\sin t \cos t}$$

$$= \frac{\sin^2 t}{\sin^2 t \cos^2 t} - \frac{\cos^2 t}{\sin^2 t \cos^2 t}$$

$$= \frac{1}{\cos^2 t} - \frac{1}{\sin^2 t}$$

$$= \sec^2 t - \csc^2 t \quad \text{Ans proved.}$$

$$\sin(\pi - \theta)$$

$$\sin\left(\pi - \frac{\pi}{4}\right)$$

LEC 30

Sum and Differences for Cosine.

$$\cos(A-B) = \cos A \cos B + \sin A \sin B$$

$$\cos(A+B) = \cos A \cos B - \sin A \sin B$$

Use difference formula to find

$$\cos(165^\circ)$$

$$\cos 165^\circ = \cos(120^\circ + 45^\circ) \text{ or } \cos(120^\circ - 45^\circ)$$

$$\cos 120^\circ = \left(-\frac{1}{2}\right)$$

$$\cos(180^\circ - 60^\circ) = \cos 60^\circ$$

$$= -\frac{1}{2}$$

$$\cos 45^\circ = \left(\frac{\sqrt{2}}{2}\right)$$

Using difference formula

$$\begin{aligned} \cos(120^\circ + 45^\circ) &= \cos 120^\circ \cos 45^\circ - \sin 120^\circ \sin 45^\circ \\ &= \left(-\frac{1}{2}\right) \left(\frac{\sqrt{2}}{2}\right) - \left(\frac{\sqrt{3}}{2}\right) \left(\frac{\sqrt{2}}{2}\right) \end{aligned}$$

$$= \frac{-\sqrt{2}}{4} - \frac{\sqrt{6}}{4}$$

$$\cos(165^\circ) = \frac{-(\sqrt{2} + \sqrt{6})}{4}$$

Sum & Difference for Sine.

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$\sin(A-B) = \sin A \cos B - \cos A \sin B$$

$$\text{find } \left(\frac{\sqrt{3}\pi}{12}\right) \sin$$

$$\sin\left(\frac{9\pi + 4\pi}{12}\right) \text{ or } \sin\left(\frac{3\pi + 10\pi}{12}\right) \quad \frac{\pi}{3} = 60^\circ$$

$$\sin\left(\frac{3\pi}{4} + \frac{\pi}{3}\right) \text{ or } \sin\left(\frac{\pi}{4} + \frac{5\pi}{6}\right) \quad \frac{\pi}{4} = 45^\circ$$

$$\sin 3\pi \sin\left(\pi - \frac{\pi}{4}\right) = \sin \frac{\pi}{4} = 45^\circ = \frac{\sqrt{2}}{2} \quad \frac{\pi}{6} = 30^\circ$$

$$\cos\left(\pi - \frac{\pi}{4}\right) = -\cos \frac{\pi}{4} = -45^\circ = -\frac{\sqrt{2}}{2} \quad \pi = 180^\circ$$

$$\sin(A+B) = \sin A \cos B + \cos A \sin B$$

$$\sin\left(\frac{3\pi}{4} + \frac{\pi}{3}\right) = \sin \frac{3\pi}{4} \cos \frac{\pi}{3} + \cos \frac{3\pi}{4} \sin \frac{\pi}{3}$$

$$= \frac{\sqrt{2}}{2} \cdot \frac{1}{2} + \left(-\frac{\sqrt{2}}{2}\right) \cdot \frac{\sqrt{3}}{2}$$

$$\sin \frac{13\pi}{12} = \frac{\sqrt{2}}{4} - \frac{\sqrt{6}}{4}$$

$$= \frac{\sqrt{2} - \sqrt{6}}{4} \text{ Ans.}$$

Sum & Difference of tangent

$$\tan A + B = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$\tan A - B = \frac{\tan A - \tan B}{1 + \tan A \tan B}$$

Exampler Find $\tan \frac{5\pi}{12}$

$$\tan \left(\frac{3\pi}{12} + \frac{2\pi}{12} \right) = \tan \left(\frac{\pi}{4} + \frac{\pi}{6} \right)$$

$$\tan(A+B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$$

$$= \frac{\tan \frac{\pi}{4} + \tan \frac{\pi}{6}}{1 - \tan \frac{\pi}{4} \tan \frac{\pi}{6}}$$

$$= \frac{1 + \frac{1}{\sqrt{3}}}{1 - (1)\left(\frac{1}{\sqrt{3}}\right)}$$

$$= \frac{\frac{\sqrt{3} + 1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}}$$

$$= \frac{\frac{\sqrt{3} + 1}{\sqrt{3}}}{\frac{\sqrt{3} - 1}{\sqrt{3}}}$$

$$= \frac{\sqrt{3} + 1}{\sqrt{3} - 1} \times \frac{\sqrt{3} + 1}{\sqrt{3} + 1}$$

$$= \frac{(\sqrt{3})^2 + 2\sqrt{3} + 1}{(\sqrt{3})^2 - (1)^2}$$

$$= \frac{3 + 2\sqrt{3} + 1}{3 - 1}$$

$$= \frac{4 + 2\sqrt{3}}{2} = \frac{2(2 + \sqrt{3})}{2} = 2 + \sqrt{3}$$

Double angle formula.

$$\textcircled{1} \sin 2A = \sin A \cos A + \cos A \sin A$$

$$= \boxed{2 \sin A \cos A}$$

$$\textcircled{2} \cos 2A = \cos (A+A)$$

$$= \cos A \cos A - \sin A \sin A$$

$$= \boxed{\cos^2 A - \sin^2 A}$$

$$\textcircled{3} \tan 2A = \tan (A+A)$$

$$= \frac{\tan A + \tan A}{1 - \tan A \tan A}$$

$$= \boxed{\frac{2 \tan A}{1 - \tan^2 A}}$$

Cosine has further 2 Formations.

$$\textcircled{4} \cos 2A = \cos^2 A - \sin^2 A$$

$$= (1 - \sin^2 A) - \sin^2 A$$

$$= \boxed{1 - 2 \sin^2 A}$$

$$\textcircled{5} \cos 2A = \cos^2 A - \sin^2 A$$

$$= \cos^2 A - (1 - \cos^2 A)$$

$$= \boxed{2 \cos^2 A - 1}$$

Example: Solve equation $\tan x \tan 2x = 1$
 give solution in interval $0^\circ < x < 180^\circ$

$$\tan x \tan 2x = 1$$

$$\frac{\tan x \cdot 2 \tan x}{1 - \tan^2 x} = 1$$

$$\frac{2 \tan^2 x}{1 - \tan^2 x} = 1$$

$$2 \tan^2 x = 1 - \tan^2 x$$

$$2 \tan^2 x + \tan^2 x = 1$$

$$3 \tan^2 x = 1$$

$$\sqrt{\tan^2 x} = \sqrt{\frac{1}{3}}$$

$$\tan x = \pm \frac{1}{\sqrt{3}}$$

$$x = \tan^{-1} \pm \frac{1}{\sqrt{3}}$$

$$x = \tan^{-1} \frac{1}{\sqrt{3}}$$

$$x = \tan^{-1} -$$

$$x = 30^\circ$$

$$, x = -30^\circ$$

$$x = n\pi + \frac{\pi}{6}$$

$$, x = n\pi - \frac{\pi}{6}$$

$$\tan x = -1 \quad \boxed{135^\circ}$$

$$0 \leq x$$

we are
with 9

So,

$$x = -$$

In 2nd
is given

$$x = 18$$

$$x \approx 15$$

Power reducing formulae

$$\cos 2A = 2\cos^2 A - 1$$

$$\cos 2A + 1 = 2\cos^2 A$$

$$\cos 2A + 1 = \cos^2 A$$

$$\textcircled{1} \cos 2A = 1 - 2\sin^2 A$$

$$\cos 2A - 1 = -2\sin^2 A$$

$$\sin^2 A = \frac{1 - \cos 2A}{2}$$

$$\tan^2 A = \frac{\sin^2 A}{\cos^2 A} = \frac{1 - \cos 2A}{\cos 2A + 1}$$

$$= \frac{1 - \cos 2A}{\cos 2A + 1}$$

Half angle identity

$$\sqrt{\frac{\sin^2 A}{2}} = \sqrt{\frac{1 - \cos 2A}{2}}$$

$$\frac{\sin A}{2} = \pm \sqrt{\frac{1 - \cos A}{2}}$$

$$\sqrt{\frac{\cos^2 A}{2}} = \sqrt{\frac{\cos 2A + 1}{2}}$$

$$\frac{\cos A}{2} = \pm \sqrt{\frac{\cos A + 1}{2}}$$

LE 31-34 (Revision).

From arithmetic to sd on-

LEC 35

Statistics

Statistic involved three activities

- Describing information (or data)
- Collecting info (or data)
- Drawing collective conclusion

Types of Statistics.

- 1- descriptive (describe the data)
2. Inferential (conclude the data)

Variable:- to measure data

Types

Quantitative age, weight

continuous

e.g weight

3.5, 3.8

Discrete

1, 2, 3, 4.

Qualitative (no numerical value need)
e.g genders

1- Population: set of possible no. in group.

e.g - total 50000 total

2. Sample: subset of population of interest

e.g we take 20000 school

36 Representation of Data

using table & graph.

Measurement Scales:

Nominal scale:-
no particular order in

qualitative scale.

e.g - religious preference, race, gender.

2. Ordinal scale:-
order wrt some characteristics.
e.g. top 10 best stores
3. Interval scale:- (quantitative)
interval b/w two variables. there is
no rational zero. e.g. temperature
4. Ratio scale:- (continuous)
it has zero point at the origin
e.g. height, age, weight.

Frequency Distribution:-

- Define interval
- Tally the observation IIII

Types of Frequency.

1. Relative frequency.

divide the f of each interval by total number.

• Commulative frequency.

it is calculated by summing the frequency starting at the lowest interval

$$\text{Frequency} = \underbrace{\text{Height} \times \text{Width}}_{\text{Frequency density}}$$

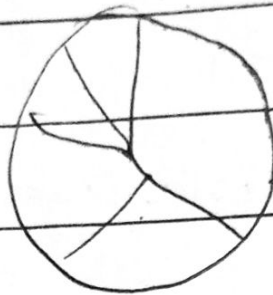
$$\text{So frequency density} = \frac{\text{Frequency}}{\text{width}}$$

Measure of central tendency.

Pie chart

it has 360° angle.

$$\frac{35 \times 100}{90}$$



$$\begin{array}{r} 37.8 \\ 9 \overline{) 380} \\ \underline{27} \end{array}$$

$$\begin{array}{r} 80 \\ 72 \\ \underline{80} \\ 72 \end{array}$$

degree

$$\frac{\text{ob. no.} \times 360}{\text{Total}}$$

to find

$$\frac{\text{ob. No.} \times \text{Total}}{360}$$

percentage

$$\frac{\text{Ob. No.} \times 100}{\text{Total no}}$$

To find number.

$$\frac{\text{ob. No.} \times \text{Total}}{100}$$

Arithmetic mean ($\bar{X}$) for ungrouped data.

$$\bar{X} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$$

For grouped data

$$\bar{X} = \frac{\sum f \cdot x}{\sum f}$$

if interval is give we take mid point
first & also make class boundaries

② Geometric Mean. (G.M)

$$G.M = \sqrt[n]{a_1 \cdot a_2 \cdot a_3 \dots a_n}$$

if the variable depends on one another then.

$$G.M = \text{antilog} \left[\frac{\sum \log x}{n} \right]$$

G.M from class interval.

$$G.M = \text{antilog} \left[\frac{\sum f \log x}{\sum f} \right]$$

Weighted Mean:-

When we have different weight.

$$\bar{X} = \frac{w_1 x_1 + w_2 x_2 + \dots + w_n x_n}{w_1 + w_2 + w_3 + \dots + w_n}$$

Harmonic mean:-

Situation involves rates & ratios

$$H.M = \frac{n}{\frac{1}{x_1} + \frac{1}{x_2} + \frac{1}{x_3} + \dots + \frac{1}{x_n}} \quad \text{or} \quad \frac{\sum f}{\sum (f/x)}$$

Median

The is the center most value of data.

$$Me = L_o + \frac{h}{f} \left(\frac{n}{2} - F \right)$$

L_o = Lower class boundary 59.5

h = width = $64.5 - 59.5 = 5$

F_o = Frequency of middle class = 29

F = Commulative frequency of previous class.

n = total weight = 98

Mode:-

value with highest frequency
or most occurring value.

Mode of grouped data.

$$M_o = L_1 + \left(\frac{\Delta_1}{\Delta_1 + \Delta_2} \right) h$$

L = Lower class boundary of modal class

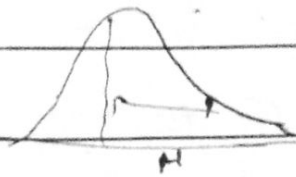
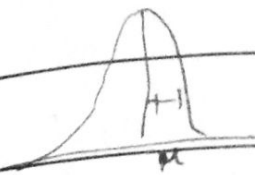
Δ_1 = Diff of frequency b/w modal class before

Δ_2 = Diff of frequency b/w modal class & after

h = width = class interval of modal class

Measure of Dispersion.

indicates scattering of data.



Range = Largest value - Smallest value.

Quartiles :- divide data into Quarters.

Quartiles = divide in 4 quarters.

Quintiles = divide in fifth parts

deciles = " " 10 parts

Percentile = " " 100 parts.

$$L_k = \frac{k}{100} (n+1)$$

Inter Quartile Range.

$$IQR = Q_3 - Q_1$$

$$\text{Average mean deviation} = \frac{\sum (x - \bar{x})}{n} = 0$$

$$\text{Mean absolute Deviation} = \frac{\sum |x - \bar{x}|}{n}$$

|MAD|

$$\text{Variance} = s^2 = \frac{\sum (x - \bar{x})^2}{n}$$

$$\text{Standard deviation} = s = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

Group data.

$$\text{Var} = s = \frac{\sum x_i f_i}{\sum f}$$

$$\text{Standard deviation} = s = \sqrt{\frac{\sum x_i f_i}{\sum f}}$$

Chebyshev's Theorem

The proportion of observation within k standard deviation

$$= 1 - \left(\frac{1}{k^2} \right)$$

Skewness & Kurtosis

Coefficient of variation:-

for two different data sets.

$$CV = \frac{S}{\bar{X}} = \frac{\text{Standard deviation of } x}{\text{Mean / average value of } x}$$

Formula :-

$$\text{Mode} = 3\text{median} - 2\text{Mean}$$

Skewness.

Positively skewed (Right Skew) more on upper tail

$$\text{mean} > \text{median} > \text{mode}$$

Negatively skewed / left skew

data with long lower tail

$$\text{mean} < \text{median} < \text{mode}$$

Coefficient of Skewness = $\frac{\text{Mean} - \text{Mode}}{\text{Standard Deviation}}$

$$= \frac{3(\text{Mean} - \text{Median})}{\text{standard deviation}}$$

Kurtosis :- (up, down)

Distribution is more or less "peaked"

than the normal.

$$B_2 = \frac{M_4}{M_2^2} = \frac{n \sum (x - \bar{x})^4}{[\sum (x - \bar{x})^2]^2}$$

Measure of Kurtosis:-

If $B_2 - 3 > 0$ then the distribution is leptokurtosis.

If $B_2 - 3 < 0$ platykurtic

If $B_2 - 3 = 0$ mesokurtic.

where 3 is kurtosis of normal distribution.